ANURAG VERMA

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Professional Summary

Python backend engineer and founding software engineer at Gynosakhi, a women's healthcare platform, where I designed and built the core backend across two services: a Django admin monolith (~50K LOC, 121 models, 385 REST endpoints) covering patient CRM, a financial/commission engine, RBAC, and notifications, plus a standalone Flask microservice powering an AI clinical copilot backed by Google Gemini. Comfortable owning systems end-to-end – schema design, business logic, and working directly with founders to scope new features, not just implement tickets. Experienced across Django, FastAPI, Flask, and MySQL/PostgreSQL, with hands-on production experience integrating LLM APIs, payment gateways, and cloud infrastructure.

Technical Skills

- **Backend Development :** Python, Django, FastAPI, Flask, RESTful API design
- **Databases :** MySQL, PostgreSQL, MongoDB – schema design, indexing, query optimization
- **System Design :** RBAC/permission engines, financial & commission-calculation engines, rules engines, scheduled jobs, notification-segmentation systems
- **AI/LLM Integration :** Google Gemini API, Anthropic Claude API – intent classification, session management, structured JSON responses
- **Cloud & Infra :** AWS S3 (boto3), Firebase Cloud Messaging, GitLab
- **DevOps & Testing :** Git, Linux, Pytest / Django TestCase
- **Data & Reporting :** Pandas, NumPy, Plotly, WeasyPrint (PDF), openpyxl (XLSX)
- **Machine Learning :** Scikit-Learn, supervised/unsupervised learning, clustering (K-Means, DBSCAN, Hierarchical)

Work Experience

Gynosakhi — Women's Healthcare Platform **Jul 2025 – Present**

Software Engineer (Founding Engineer)

AI Clinical Copilot — Flask microservice, Google Gemini 2.5 Flash

- Designed and built a standalone Flask API (separate from the Django admin monolith) powering a conversational AI assistant for doctors, exposing REST endpoints for chat, session management, and live patient search.
- Built a keyword-based intent-classification layer routing each query into patient-specific, doctor-context, or general-medical buckets before querying the database – avoiding unnecessary lookups.
- Built a context service that fetches only the MySQL data each intent needs and rebuilds the LLM prompt fresh from the database on every request, keeping the assistant stateless and accurate.
- Built a thread-safe, in-memory session manager with automatic eviction of sessions idle over 30 minutes, and a structured JSON response contract so the frontend renders tables/charts independently of the conversational reply.

Financial & Commission Engine

- Built a 2,200+ line profit-and-loss engine computing gross revenue → commission → clinician payouts → refunds → GST → net profit, with unified entity profiling across doctors, nurses, labs, service plans, and organizations.
- Designed a 3-tier commission-resolution cascade (direct match → doctor/service-type lookup → fallback split) with ratio-scaled payouts so coupons/discounts never break agreed commission splits.

RBAC & Access Control

- Designed a 4-tier permission-resolution engine (superuser bypass → user-level deny → user-level allow → role default) covering 35+ modules across 6 roles (~140 permission codes).
- Built signal-driven permission lifecycle management so adding or removing an admin module automatically creates/cleans up its permissions with zero manual work.

Patient CRM, Analytics & Automation

- Designed 12+ interconnected models covering pregnancy tracking, a fertility (TTC) cycle-logging subsystem, care-team assignment, and a follow-up CRM with intent scoring (cold → hot → converted).
- Built a Django cron job that identifies patients completing a pregnancy week each day via a raw-SQL day-count query, pulls personalized health data from an external API, and sends templated weekly-update emails with duplicate-send protection.
- Built a Firebase Cloud Messaging engine with 20+ audience segments, 500-token batched multicast delivery, and automatic stale-token cleanup.
- Designed and optimized the MySQL schema underlying the platform (121 models / ~65 tables) and wrote Pytest/Django TestCase coverage across views, models, forms, and notification logic.
- Collaborate directly with founders and the mobile app team to scope, prioritize, and ship new features end-to-end.

Zefrokart — E-commerce Platform **Jun 2024 – Jun 2025**

Software Developer

- Built and maintained an admin panel covering product, order, and user management for the platform.
- Designed and optimized database schemas for inventory, customer data, and sales records.
- Built RESTful APIs connecting the admin panel to the user-facing storefront, with security measures to protect sensitive data.

Selected Projects

Gynosakhi Copilot (Beta) — Role: Backend/API Engineer

- **Technologies:** Python, Flask, Google Gemini API, REST, real-time data injection
- Built the API layer for a doctor-facing AI assistant answering natural-language questions grounded in real clinical and business data – e.g. earnings this month, patient counts, or a specific patient's latest vitals – alongside general medical Q&A.

AI Call Agent (Ongoing Research) — Role: Backend Developer & AI Integrator

- **Technologies:** Twilio, Google Dialogflow, Google TTS, Python Flask
- Designing a 24×7 multilingual voice assistant for patient queries, appointment bookings, and emergency routing, with planned sentiment analysis and call recording.

Real Estate Marketplace — Role: Full-Stack Developer (Internship)

- **Technologies:** MongoDB, Express.js, Node.js, React, Redux Toolkit, JWT
- Built a full-stack real estate marketplace (MERN) with JWT authentication and Redux Toolkit state management. Live: anurag-stack.onrender.com

WaveTranscribe — Role: Backend Developer

- **Technologies:** FastAPI, Whisper AI, Matplotlib, PyAudio, Pydub
- Built a speech-to-text service using Whisper AI with FastAPI endpoints for upload, real-time transcription, and live waveform visualization.

Survey Application — Role: Backend Developer

- **Technologies:** FastAPI, PostgreSQL, Python
- Designed a PostgreSQL schema and REST APIs for survey creation, response submission, and result analysis, with authentication and validation.

Blog Platform (Pen&Paper) — Role: Backend Developer

- **Technologies:** Django, HTML, CSS, Bootstrap
- Built an admin-authored blog platform with full post management and public read/comment access.

Education

Dr. APJ Abdul Kalam Technical University (AKTU) **2024**
MCA – Computer Applications ***CGPA 7.92***

Veer Bahadur Singh Purvanchal University, Jaunpur **2021**
B.Sc (Math)

Achievements

- **5-Star** rating on HackerRank in Python, demonstrating advanced problem-solving skills.
- **4-Star** rating on HackerRank in C, demonstrating proficiency in C programming and algorithmic challenges.
- Completed the **#100DaysOfCode** challenge, mastering Python and advancing in Data Science.

Certificates

- Python for Data Science – (IBM)
- Applied Data Science with Python – Level 2 – (IBM)
- Machine Learning with Python – Level 1 – (IBM)
- Python and Django Full Stack Web Developer Bootcamp – (Udemy)
- 100 Days of Code: The Complete Python Pro Bootcamp – (Udemy)